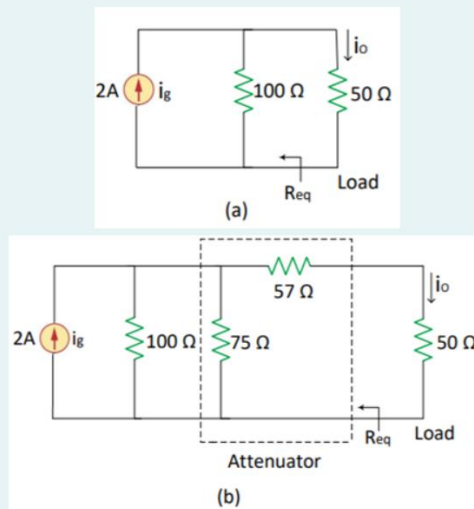
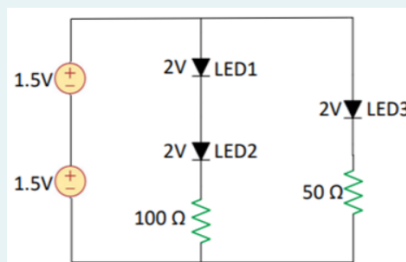


QUESTION 1 [17 marks]

- a. (8 marks) An attenuator is an interface circuit that reduces the voltage or current level without changing the output resistance. For the circuit shown in Figure 1, calculate:
- (2 marks) Output current (i_o) in the circuit without attenuator (Figure 1(a)).
 - (4 marks) Output current (i_o) in the circuit with attenuator (Figure 1(b)).
 - (2 marks) Equivalent resistance (R_{eq}) of the circuit with attenuator (Figure 1(b)) when the current source is turned off ($i_g = 0A$).



- b. (9 marks) The circuit shown in Figure 2 is used to test an LED torch design. The torch is powered by AA batteries (1.5 V each) and has white LEDs with forward voltages across their terminals equal to 2 V. The maximum current that can flow through the LEDs is 25mA. For currents higher than that, the LEDs get damaged and do not light up.
- (6 marks) Which LED(s) will light up in the torch? Please, indicate the LED(s) name(s) clearly and support your answer with the appropriate calculations.
 - (3 marks) Find the power dissipated in the 50 Ω resistor.



QUESTION 2 [20 marks]

a. (10 marks) In the circuit shown in Figure 3,

i. (3 marks) Find the voltage V_{ab} .

ii. (7 marks) Calculate all the powers absorbed and/or supplied by the 5 elements and verify the law of conservation of energy.

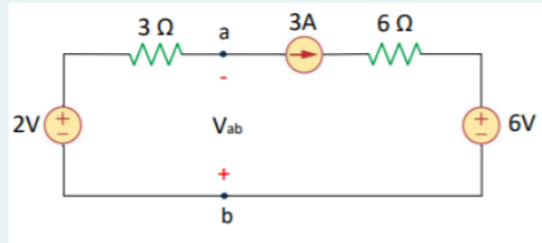


Figure 3

b. (10 marks) For the circuit shown in Figure 4, apply mesh analysis and write down the mesh equations using the labels provided.

Note: Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a , b , c and d can be positive or negative numbers, but **DO NOT** solve them.

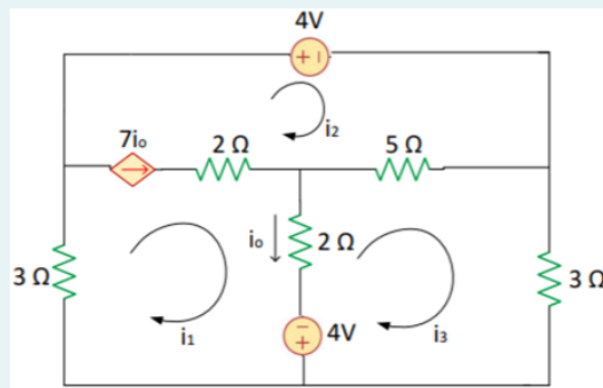


Figure 4

QUESTION 3 [18 marks]

The circuit in Figure 5 corresponds to a bicycle with front and rear bulb lights and a permanent magnet generator (hub dynamo) and battery to power them. The circuit also includes a USB charger to charge the mobile phone while riding.

When the bicycle is travelling at 20km/h (average cycling speed), the permanent magnet generator voltage is 8 V.

- (15 marks) Use nodal analysis to calculate the current flowing through the generator when the bicycle is travelling at 20 km/h and the phone charger is connected (switch Sw1 is closed).
- (3 marks) Calculate the power dissipated by the 20 Ω resistor.

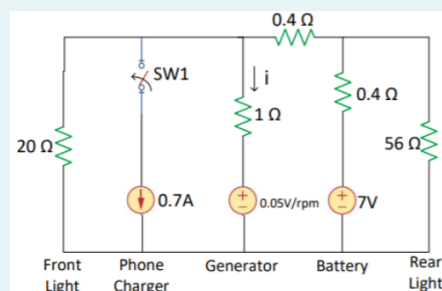


Figure 5

QUESTION 4 [30 marks]

a. (12 marks) For the circuit shown in Figure 6, use a succession of source transformations (only source transformations and resistor/source simplification) to find an equivalent circuit consisting of a single current source and a single resistor, as seen from terminals a - b . Then, determine the Norton current, I_N , and Norton resistance, R_N , at the terminals a - b .

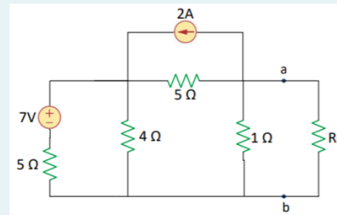


Figure 6

b. (18 marks) The circuit shown in Figure 7 is being used to power a water heater. The water heater can be modelled as a single resistor connected to terminals a - b .

- (8 marks) Calculate the Thevenin voltage at terminals a - b .
- (8 marks) Calculate the water heater resistance that will ensure the maximum transfer of power from the circuit to the heater.
- (2 marks) Find the maximum power that can be delivered to the heater.

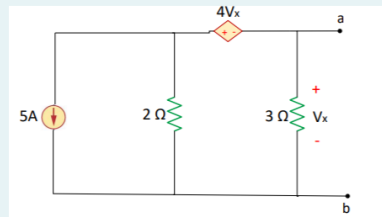


Figure 7

QUESTION 5 [15 marks]

Switch SW1 in the circuit shown in Figure 8 was open for a long time. It closed at $t = 0$ s and, after remaining closed for 3 ms, it opened again.

- (9 marks) Calculate $v_o(t)$ for $0 \leq t < 3$ ms.
- (6 marks) Calculate $v_o(t)$ for $t \geq 3$ ms.

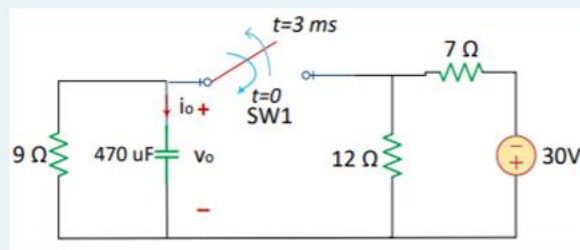


Figure 8